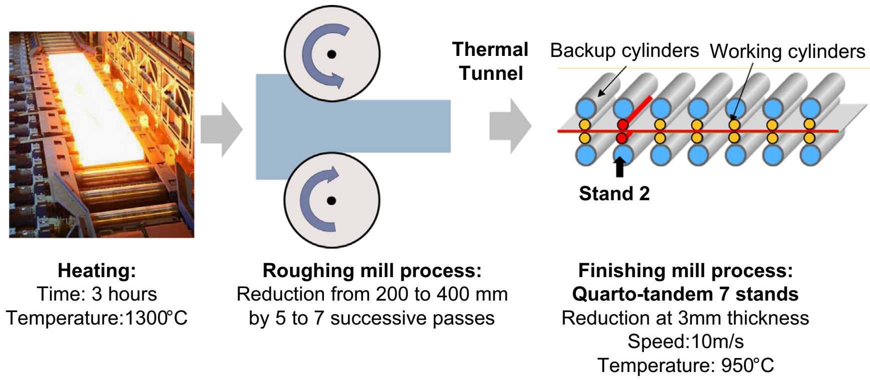

Fig. 1. Hot rolling mill process description.

explained by the same authors to be linked with product temperature. Depending on the grade, sticking temperature vary between 900 °C for low alloyed grades (such as 430) up to 1070 °C for high alloyed grades (such as 436). A second order rolling parameter pointed out in the literature is the rolling schedule (i.e. thickness reduction ratio) as described in [5]. Indeed, local high stresses will generate more fractures on the product's oxide layer, therefore generating more particles onto the working roll surface. A detailed study on the finishing mill reveals that the second stand was particularly inclined to generating "rolled-in scale" defect rather than other stands, because rolling temperatures on product edges were, for some sensitive grades, equal to the sticking temperature. Moreover, the rolling pass schedule was adapted to reduce the severity of the defect in the sense of decreasing rolling pressure on stand 2. That is why, we will investigate the wear evolution on working rolls used in the second stand with a special focus on product edges' contact areas.

This study will be focused on austenitic alloys and working rolls made of HSS (High strength steel). Consequently, working cylinders from this stand will be studied at different kilometers of rolled strips to characterize their wear. Austenitic stainless steel sheet have been studied.

2.2. Experimental design

In order to characterize the wear of hot rolling mill cylinders during the finishing stage, different sections of the cylinder's surface are chosen according to different process parameters as described in Table 1.

The cylinder's surface sections have been identified according to the kilometer of the cylinder, the position along the cylinder's length (11 axial positions between 00 and 20, from "the motor side" to "the operator side"), the vertical location of the working cylinders (inferior or superior), and the generatrix number (between 1 and 6). Fig. 2 shows all replicas' positions on the hot rolling milled cylinders. As a result, it enables us to analyse the evolution of roughness of rolled strip between 0 and 149 km.

2.3. Replication

Replicas are made on the working cylinders's surface on the 2nd stand of the finishing mill process following a defect at this step of the manufacturing process. Because of the large dimensions of the working rolls, from 680 mm to 760 mm in diameter and 2500 mm in length, some replicas are made on the surface of working cylinders. Roughness measurements are therefore readily accessible to usual laboratory measuring devices. Replicas allow us to obtain the negative of the surfaces to duplicate the surface's roughness. The Repliset T3® resin (proposed by Struers®) is used to duplicate surfaces after cleaning the cylinder with ethanol. In

Table 1
Hot rolling mill process parameters used in order to characterize wear of working cylinders.

Stainless steel type	Austenitic
Working rolls kilometers	0, 51, 88, 149 km
Working rolls location	Inferior; superior
Lateral location	From operator side to motor side in 11 positions with equidistance
Working rolls homogeneity	6 generatrix

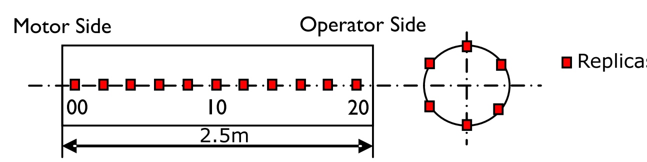

Fig. 2. Position of the replicas on hot rolling mill cylinders.

order to characterize the damage on the working cylinders' surfaces, replicas are made at different kilometers of rolled strips between two grinding operations.

2.4. Roughness measurements

The white light interferometer (NewView 7300, Zygo™) is used to measure specimen topography with an $\times 50$ Mirau objective, 0.52 µm lateral resolution and 0.01 µm vertical resolutions are obtained. The inspected surface area is 0.96 mm by 0.96 mm acquired by stitching on each measurement. The topographical image size is 2176×2176 pixels. Five surfaces are randomly measured on each section of the cylinder surfaces analyzed in order to obtain robust statistic results.

2.5. Measurements treatment

To get reliable physical interpretation, it is very important to carry out a multi-scale analysis. Consequently, before using any filter, treatments have to be applied on all measurements. Firstly, a vertical plane symmetry perpendicular to the z axis is applied. This treatment allows us to re-establish the direction of the surface roughness altitudes. Then, a polynomial fitting is applied to remove the measured surface's form. A 3rd degree polynomial fitting is used to delete the surface form. Table 2 allows to observe surfaces after applying a polynomial treatment at different degrees. 1st and 2nd degree polynomial fittings are not adapted to the measured surfaces because they are neither flat (1st degree) nor symmetrical (2nd degree). If the polynomial degree exceeds 3, the surface's waviness might be damaged by the treatment (Table 2). When a 5 degree polynom is applied, the waviness starts